

LAB 02 – PROMPT HISTORY

Reservoir Stage Analysis · Numerical Methods · BES6-M

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AI Assistant: ChatGPT (OpenAI)

1. INITIAL REQUEST – Understanding the Assignment

User Prompt:

"tell me what i need to do"

AI Response:

Provided a complete breakdown of the lab activity:

- Understanding the data (96 readings, 15-minute intervals)
- Reusing previous work (finite-difference derivatives)
- Choosing and defending a model (logistic, Gompertz, sum of logistics, or cubic)
- Running Levenberg-Marquardt fit with proper initial guesses
- Computing statistics (SSE, R^2 , s, parameter errors, t, p-values)
- Integrating the fitted curve

- Generating an HTML dashboard with all numbers from Python
 - Submitting 4 files to GitHub
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2. REQUEST FOR CODE GUIDANCE

User Prompt:

- Reusing previous work (finite-difference derivatives)
- Understanding the data (96 readings, 15-minute intervals)
- Choosing and defending a model (logistic, Gompertz, sum of logistics, or cubic)
- Running Levenberg-Marquardt fit with proper initial guesses
- Computing statistics (SSE, R^2 , s, parameter errors, t, p-values)
- Integrating the fitted curve
- Generating an HTML dashboard with all numbers from Python

AI Response:

Provide a complete solution and offer guidance and a detailed skeleton template.

3. DATA FILE PROVIDED – Excel Loading

User Prompt:

"Here is the excel file"

AI Response:

Provided data loading code:

```
python
```

```
df = pd.read_excel('Data 01.xlsx', sheet_name='Sensor Log', skiprows=3)
df = df.head(96).copy()
df['Timestamp'] = pd.to_datetime(df['Timestamp'])
t0 = df['Timestamp'].iloc[0]
df['hours'] = (df['Timestamp'] - t0).dt.total_seconds() / 3600.0
t = df['hours'].values
```

```
h = df['Depth (m)'].values
```

4. ERROR DEBUGGING – FileNotFoundError

User Prompt:

"Traceback (most recent call last): ... FileNotFoundError: [Errno 2] No such file or directory: 'Data 01.xlsx'"

AI Response:

Fixed by adding robust file-finding logic and absolute path support with multiple candidate paths.

5. ERROR DEBUGGING – NumPy trapz Deprecation

User Prompt:

"AttributeError: module 'numpy' has no attribute 'trapz'. Did you mean:
'trace'?"

AI Response:

Replaced `np.trapz` with `trapezoid` from `scipy.integrate`.

6. ERROR DEBUGGING – Unicode Encoding

User Prompt:

"UnicodeEncodeError: 'charmap' codec can't encode character '\u2011' ..."

AI Response:

Fixed by adding `encoding='utf-8'` when writing the HTML file.

7. SUCCESSFUL RUN

User Prompt:

"Loading data from: c:\Users\Jayson\OneDrive\Desktop\deliverables\Data
01.xlsx Dashboard generated: lab02_surname.html Max dh/dt: 0.120000
m/h at t = 0.00 h SSE = 0.018600, R² = 0.977441, s = 0.014219 m"

AI Response:

Confirmed successful execution and provided next steps for submission.

8. LOCATING THE HTML FILE

User Prompt:

"where did the html got saved?"

AI Response:

Explained the file was saved in the script directory and provided
instructions to locate it.

9. DESIGN REQUEST – Futuristic Theme

User Prompt:

"make it eye appeasing"

AI Response:

Redesigned the dashboard with:

- Glass-morphism container
 - Gradient header with overlay
 - Hover animations on tabs and cards
 - Professional color palette
 - Responsive layout
 - Stat cards with colored borders
-

10. DESIGN REQUEST – Futuristic with Animation

User Prompt:

"make the design look like futuristic"

AI Response:

Created dark theme with:

- Animated grid background
 - Neon cyan/magenta accents
 - Glowing text shadows
 - Glass-panel effects with backdrop blur
 - Floating animations
-

11. TAB EXPLANATIONS

User Prompt:

"explain each tab"

AI Response:

Provided detailed explanation of each dashboard tab:

1. Summary – Key findings and overview
 2. Data Overview – Dataset description and structure
 3. Derivatives – First and second derivatives
 4. Fit & Residuals – Model validation
 5. Statistics – Parameter table and goodness-of-fit
 6. Area – Numerical integration results
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12. ERROR DEBUGGING – f-string Braces

User Prompt:

"SyntaxError: f-string: expecting '=', or '!', or ':', or '"'" (at line 820)

AI Response:

Fixed by doubling all curly braces in CSS and JavaScript (`{{` and `}}`) so Python treats them as literal characters while keeping single braces for actual data placeholders.

13. FINAL FIX – HTML Display Issues

User Prompt:

"fix, i cant click the tab and the images dont show"

AI Response:

Corrected all remaining escaped brace issues in:

- CSS rules (all { → {})
 - JavaScript object literals and conditionals
 - The showTab function with properly escaped map and if blocks
-

End of Prompt History
